Deep learning from the Palisades and Eaton fires using a novel geospatial dataset

Arye Mindell, Tinbite Ermias, Lei Huang
Department of Electrical and Computer Engineering
Loyola Marymount University

INTRODUCTION

The Challenge: Accurately predicting wildfire progression in real-time, especially during extreme weather events, remains a major challenge. Traditional models rely heavily on physical simulations that are computationally expensive and often struggle to incorporate complex and heterogeneous data in a timely manner.

The Fires: The Palisades and Eaton fires (Jan 2025) serve as critical case studies due to their rapid and erratic spread patterns, influenced by severe wind, steep slopes, and proximity to the Wildland-Urban Interface (WUI).

The Research Problem: To develop a highly efficient deep learning framework that integrates multi-source geospatial and temporal data to predict dynamic fire spread maps with high accuracy.

OBJECTIVES

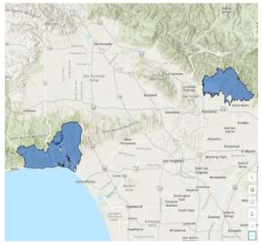
There are two main objectives of this project

1. **Data Collection**: to collect and curate a novel time-series geospatial dataset that captures the dynamic behavior of these fires over time and space.
2. **Model Development**: to train and evaluate deep learning models capable of identifying patterns and relationships that influence fire spread.

METHOD OF DATA COLLECTION

Tools used:

- ArcGIS
- Google Earth Engine API



(a) Mapping of the Palisades and Eaton Fires

Data Processing:

- Feature integration
- Data normalization

(b) VIIRS Fire map for Palisades Fire

(c) VIIRS Fire map for Eaton Fire

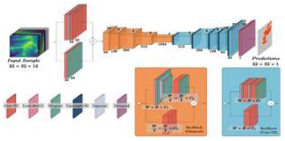
Figure 1. Palisades and Eaton Fire Data Examples

DATA SOURCES

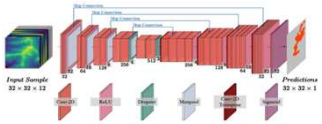
Table 1. Data sources and features

Category	Provider	Dataset	Features
Fire data	CAL FIRE	California Fire Perimeters	Spatial and temporal information of fires
	NASA	MOD14A1.061	Daily global fire mask derived from MODIS thermal radiances
	NASA/NOAA	VJ114IMGTDL_NRT	Brightness temperatures, confidence levels, and fire radiative power
Weather data	UC Merced	GRIDMET	Temperature, precipitation, winds, humidity, radiation, and drought indices
Vegetation	NASA	VNP13A1.002	VIIRS vegetation indices including NDVI, EVI, and EVI2
Elevation	NASA/USGS	SRTMGL1.003	Digital elevation at a 30m resolution
Population	NASA	CIESIN_SEDAC_GPWv4_POPDENS_R11	Human population estimates at a 30 arc-second resolution

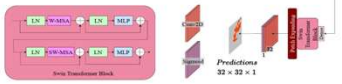
MODELS



(a) Autoencoder



(b) UNet



(c) Transformer-based SWIN-UNet

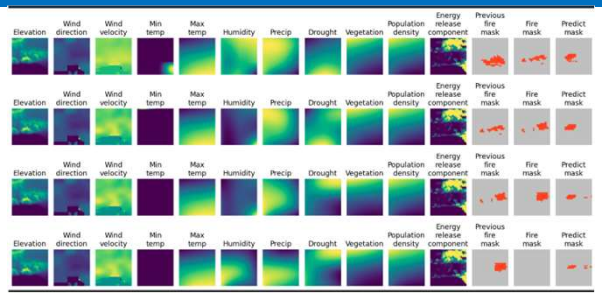
Figure 2. Deep Learning Models Tested [2]

RESULTS

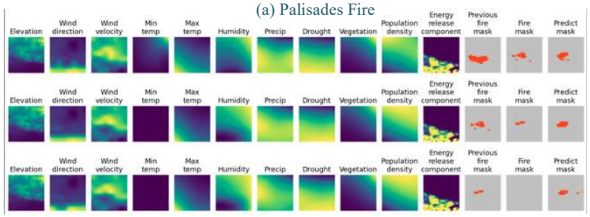
Table 2. AUC-PR performances on baseline and new testing datasets

Datasets	Models		
Fire Event	Autoencoder	U-Net	Swin U-Net
Baseline[1]	0.3330	0.2804	0.3376
Eaton	0.2422	0.0870	0.2923
Palisades	0.0867	0.1027	0.1426

VISUALIZATION



(a) Palisades Fire



(b) Eaton Fire

Figure 3. Visualized predictions

CONCLUSIONS

A comprehensive multi-modal spatial-temporal dataset has been created from the recent Palisades and Eaton Fires. It integrates multi-temporal features, including dynamic weather, vegetation moisture, and fire progression data, alongside critical static information like elevation and population density. Preliminary evaluations using existing deep learning models demonstrate its utility and highlight avenues for improvement. This dataset lays the foundation for developing more advanced models that can deliver the accuracy and efficiency needed for actionable, real-time fire management under severe weather conditions.

FUTURTE WORK

1. **Data Collection**
   - Automating data collection from other historical fires
   - Collecting features with enhanced spatial and temporal resolutions
2. **Model Development**
   - Improving the existing transformer-based model
   - Developing new hybrid models exploiting spatial propagation mechanisms (GNNs) and the long-range temporal dependencies (Transformers)

REFERENCES

[1] F. Huot, et al. "Next day wildfire spread: A machine learning dataset to predict wildfire spreading from remote-sensing data." *IEEE Transactions on Geoscience and Remote Sensing,* 60, pp.1-13, 2022.

[2] Y. Zhou, et al. "Comparative and interpretative analysis of CNN and transformer models in predicting wildfire spread using remote sensing data." *Journal of Geophysical Research: Machine Learning and Computation* 2.2, p.e2024JH000409, 2025